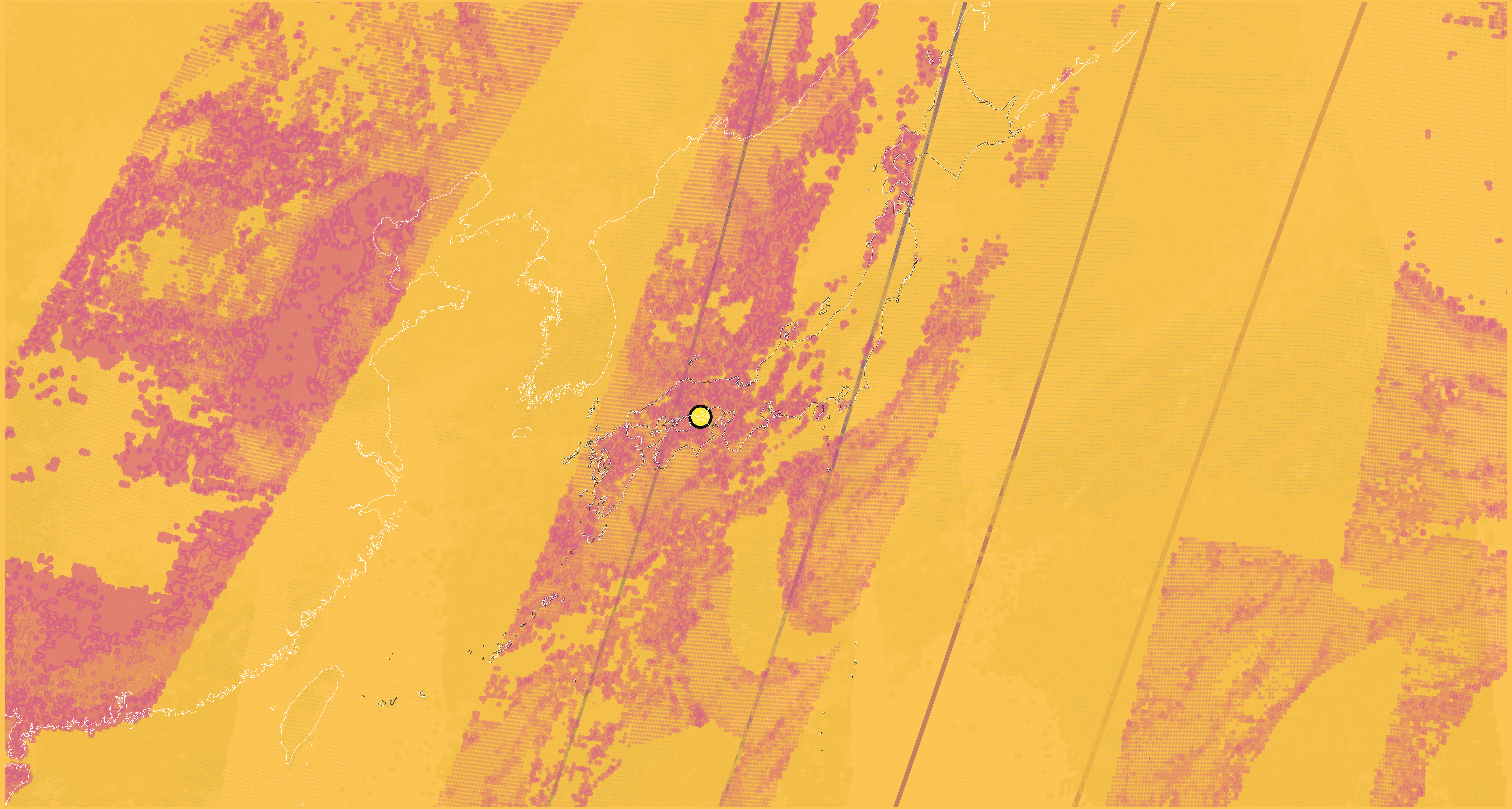


Data bibliography

Atmosphere / Japan

01 Hajime Narukawa — AuthaGraph formulation
2022 formulation; native Cartofreako implementation
Projection and graticules are mathematical constructions. Horizontal periodicity supports the ST2 framing study.
https://doi.org/10.11212/jjca.60.1_1

02 Natural Earth — 1:10m physical vectors
Pinned local vectors used by the 29 October 2026 delivery; not new observations
Land, coastline, minor islands, lakes, rivers, ice and generalized bathymetry. Individual version files and component hashes are in the JSON ledger.
https://www.naturalearthdata.com/downloads/10m-physical-vectors/
Attribution / use: Public domain

03 JAXA P-Tree Himawari-9 Level-2 Cloud Property 1.0
29 October 2026, 18:20 UTC
Himawari full disk; 10-minute, approximately 5 km, daytime retrieval only
ftps://ftp.ptree.jaxa.jp:990/pub/himawari/L2/CLP/010/
Attribution / use: Provider attribution and reuse conditions: https://www.eorc.jaxa.jp/ptree/terms.html

04 GCOM-C/SGLI daily daytime aerosol optical depth at 500 nm
15 October 2026, 00:00 UTC
global daily observed aerosol column loading; not smoke attribution or PM2.5 exposure
https://data.earth.jaxa.jp/stac/cog/v1/catalog.json
Attribution / use: Provider attribution and reuse conditions: https://earth.jaxa.jp/en/data/policy/

05 JAXA GSMaP daily gauge-adjusted precipitation rate
26 October 2026, 23:00 UTC
60S to 60N daily precipitation-rate field in mm h-1; not flood or extreme-event counts
https://data.earth.jaxa.jp/stac/cog/v1/catalog.json
Attribution / use: Provider attribution and reuse conditions: https://earth.jaxa.jp/en/data/policy/

06 JASMES Aqua/MODIS daily surface shortwave radiation
15 October 2026, 23:59 UTC
global daily surface shortwave radiation with publication lag
https://data.earth.jaxa.jp/stac/cog/v1/catalog.json
Attribution / use: Provider attribution and reuse conditions: https://earth.jaxa.jp/en/data/policy/

07 Natural Earth — Japan geographic boundary
1:10m boundary extracted for the 29 October 2026 delivery
Defines the Japan/Setouchi viewport. It does not turn national statistics into island-level measurements.
https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-0-countries/
Attribution / use: Public domain

Measure and interpretation
Physical observations / 15–29 October 2026
H3 resolution 8; per-layer quality masks and valid-sample threshold retained. Source times differ. Solar geometry is calculated at 29 October 2026, 20:00 UTC. Land fill opacity zero, outlines above data.

H3r8 resolution trial · original observation dates and sampling retained

Actual H3 cell boundaries; original source-center values reaggregated directly at this resolution.

Unoccupied cells have no sampled center: unobserved, not zero. Source-pixel footprints are not filled.

Finer indexing adds no observations. Data layers render at 300 dpi; PDF embeds them. Outlines and type remain vector.

Atmosphere: maximum 720 samples per raster axis, quality masks retained; radiation uses the same 100 opacity levels.

Japan companion
Same evidence in the Japan/Setouchi geographic viewport; neighboring countries may appear. No island-level value is inferred. The nighttime snapshot contains no positive daytime cloud retrievals; missing cloud is not clear sky.

Source ledger
Pinned input files, complete source manifests and checksums:
https://bdekoz.github.io/cartofreako/docs/pages/gallery/setouchi-2028-bibliography.json

Japan / Solar / cloud
Atmosphere

Physical observations / 15–29 October 2026
Land outlines above data; no land fill. Missing observations are not zero.

World dataset: JAXA/EORC observations; NASA MODIS shortwave input.

- Shortwave radiation · 0–400 W/m² · 100 opacity bands · 15 Oct ●
- Aerosol optical depth · 500 nm · 15 Oct ●
- Precipitation · mm/hour · 26 Oct ●
- Confident-cloud fraction · 29 Oct ●
- Cloud optical thickness · 29 Oct ●
- Cloud-top height · km · 29 Oct ●
- ISCCP cloud type · 29 Oct ●

Cloud retrievals are regional and daytime only. AOD is not smoke or PM2.5.

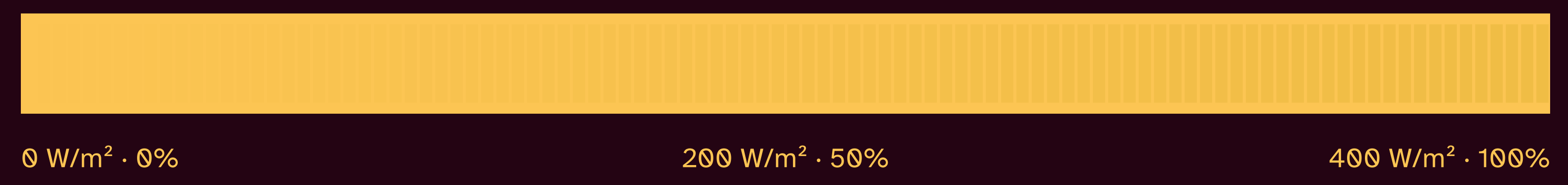
Cartographic image width: 43.5 inches on a 44-inch sheet; artwork scaled proportionally to 98.863636%.

Japan detail includes neighboring countries; it is not a Japan-only total.

Yellow point: conceptual Setouchi reference (34.45°N, 133.99°E); venue unconfirmed.

Radiation opacity · 100 bands

Stronger cells are more opaque; the same scale applies to both views.



Low values fade to transparent. Missing observations are not zero.

Data bibliography

Temperature 2026 / Japan

01 Hajime Narukawa — AuthaGraph formulation
2022 formulation; native Cartofreako implementation
Projection and graticules are mathematical constructions. Horizontal periodicity supports the ST2 framing study.
https://doi.org/10.11212/jjca.60.1_1

02 Natural Earth — 1:10m physical vectors
Pinned local vectors used by the 29 October 2026 delivery; not new observations
Land, coastline, minor islands, lakes, rivers, ice and generalized bathymetry. Individual version files and component hashes are in the JSON ledger.
https://www.naturalearthdata.com/downloads/10m-physical-vectors/
Attribution / use: Public domain

03 NOAA CPC / NOAA PSL — Global Unified Temperature V1.0
2025 complete; 2026 through 28 October
0.5° land TMAX/TMIN analysis aggregated to H3 resolution 8. Same-calendar-day strict records with at least 30 valid history years.
https://psl.noaa.gov/data/gridded/data.cpc.globaltemp.html
Attribution / use: No usage restrictions; NOAA PSL acknowledgment requested.

04 Natural Earth — Japan geographic boundary
1:10m boundary extracted for the 29 October 2026 delivery
Defines the Japan/Setouchi viewport. It does not turn national statistics into island-level measurements.
https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-0-countries/
Attribution / use: Public domain

Measure and interpretation
Days per H3 cell / 2026 through 28 October
62,881 H3 resolution 8 land cells. Strict same-calendar-day maximum and minimum records against 1979–2025, with ≥30 valid history years. 2026 is partial; counts are not a year-on-year warming estimate.

Japan companion
Same evidence in the Japan/Setouchi geographic viewport; neighboring countries may appear. No island-level value is inferred.

Source ledger
Pinned input files, complete source manifests and checksums:
https://bdekoz.github.io/cartofreako/docs/pages/gallery/setouchi-2028-bibliography.json

H3r8 resolution trial · original observation dates and sampling retained

Actual H3 cell boundaries; original source-center values reaggregated directly at this resolution.

Unoccupied cells have no sampled center: unobserved, not zero. Source-pixel footprints are not filled.

Finer indexing adds no observations. Data layers render at 300 dpi; PDF embeds them. Outlines and type remain vector.

Temperature: 0.5° CPC centers; daily spatial means precede strict record comparisons. Counts are not spatially summed.

Days per H3 cell / 2026 through 28 October

Six Yellow–red count classes. Neutral observed cells: zero; blank: no valid sample

World dataset: 62,881 covered cells; NOAA CPC land-temperature analysis.

81+	🟥
41-80	🟥
21-40	🟧
11-20	🟧
6-10	🟨
1-5	⬜

Strict same-calendar-day records; at least 30 years of history.

Blue outlines show record-low days. Partial-year counts are not annual trends.

Cartographic image width: 43.5 inches on a 44-inch sheet; artwork scaled proportionally to 98.863636%.

Japan detail includes neighboring countries; it is not a Japan-only total.

Yellow point: conceptual Setouchi reference (34.45°N, 133.99°E); venue unconfirmed.